

FinTrust Well-Architected Review

Week 11 pillar map | start with the business outcome, then choose the service

Operational Excellence Question: Are operations run and improved effectively? Look for: runbooks, deployments, drift FinTrust direction: Move legacy RDS configuration into reviewed CloudFormation.	Security Question: Are data, systems and identities protected? Look for: least privilege, audit, encryption FinTrust direction: Use Session Manager and Secrets Manager to reduce direct access.
Reliability Question: Can the workload recover and meet demand? Look for: AZ failure, RTO, RPO FinTrust direction: Remediate Single-AZ databases and preserve the 45 minute recovery target.	Performance Efficiency Question: Are resources meeting demand efficiently? Look for: low latency, cache, traffic spike FinTrust direction: Cache repeated product reads with ElastiCache for Redis.
Cost Optimisation Question: Is spend necessary for the stated requirement? Look for: avoid waste, interruption tolerated FinTrust direction: Use Spot for restartable overnight batch work.	Sustainability Question: Is environmental impact being reduced? Look for: energy, utilisation, carbon FinTrust direction: Test Graviton for compatible workloads and measure the result.

Use the dominant requirement as the tie-breaker. A service can support several pillars, but the scenario tells you which outcome matters most.

Supplied FinTrust Scenarios

Each row is mapped to its primary concern. The mapping is a course exercise, not a completed AWS Well-Architected Tool export.

Three RDS instances are manually configured.	Operational Excellence	Manual changes create drift and weaken repeatable operations.	Use CloudFormation, change sets and drift detection.
Admins need audited access and rotating database passwords.	Security	The goal is to reduce unauthorised direct access to sensitive data.	Use Session Manager, Secrets Manager and CloudTrail.
Two legacy MySQL RDS instances are Single-AZ; RTO is 45 minutes.	Reliability	Availability Zone failure and recovery time are the stated concerns.	Remediate Single-AZ risk and retain pilot-light recovery.
Repeated product reads take 80ms and need to be below 5ms.	Performance Efficiency	Latency under repeated load is the measure of success.	Use an ElastiCache for Redis read path.
A restartable nightly batch workload still uses On-Demand EC2.	Cost Optimisation	Interruption is tolerated, so spend can fall without breaking the need.	Use Spot capacity with checkpointing and restart handling.
Compute must reduce energy use and cost with no code change.	Sustainability	Environmental impact is explicit; Graviton also supports cost reduction.	Test a compatible Graviton family with workload metrics.

Useful distinction: an Auto Scaling Group that replaces an unhealthy instance is a Reliability answer. The same Auto Scaling Group responding to a traffic spike is a Performance Efficiency answer.

Decision Route and Review Priorities

A practical reading route for Well-Architected questions and the Week 11 FinTrust findings.

1. Read the outcome
Availability or recovery? Reliability. Access or audit? Security. Latency or scale? Performance. Spend? Cost. Environmental impact? Sustainability. Deployment process or runbooks? Operational Excellence.
2. Check the constraint
A permitted interruption opens the door to Spot. A strict payment SLA makes Reliability dominant. A stated retrieval window decides whether archival storage is suitable.
3. Choose the smallest fitting change
Select the service that addresses the named need before adding optional complexity. Record the trade-off, then validate it with real workload evidence.

Reliability and security first	Single-AZ databases and direct privileged access are risks before cost savings.	Authorised account configuration or review output.
Then cost controls	Spot and archive classes depend on explicit interruption and retrieval requirements.	Measured workload and retrieval evidence.
Improve reporting confidence	Tag coverage has to improve before chargeback can be trusted.	Tag compliance report and Cost and Usage Report analysis.

This diagram is a learning and review aid. It does not represent a completed AWS review, measured cost saving or production change.